

INTRODUCTION TO DATA SCIENCE

Data Fundamentals

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- 1 Data Science Basic Concepts
- 2 What is to be a Data Scientist



Outline

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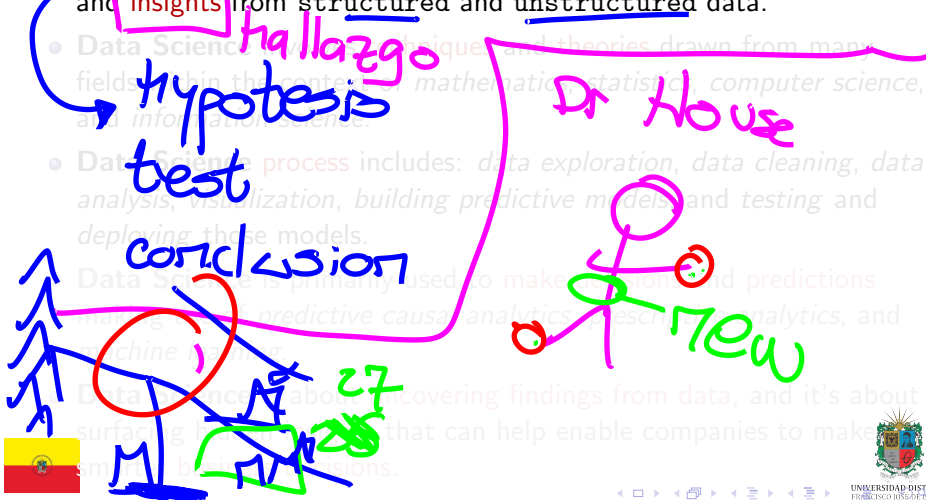


History of Data



What is Data Science?

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- **Data Science** involves **techniques** and **theories** drawn from many fields within the context of *mathematics, statistics, computer science, and information science*.
Handwritten notes: "computational" above "techniques", "math → science" above "theories".
- **Data Science** includes: *data exploration, data cleaning, data analysis, visualization, building predictive models and testing and deploying those models*.
Handwritten notes: A red box highlights "building predictive models and testing and deploying those models". A blue arrow points from "Data Science" to this box. A red arrow points from the box to "250 terabytes".
- **Data Science** is primarily used to make decisions and predictions making use of *predictive causal analytics, prescriptive analytics, and machine learning*.
Handwritten notes: A red box highlights "predictive causal analytics, prescriptive analytics, and machine learning". A blue scribble is next to it. A red arrow points from the box to "250 terabytes".
- **Data Science** is about uncovering findings from data, and it's about surfacing insights that can help companies to make smarter business decisions.
Handwritten notes: A green box highlights "uncovering findings from data". A green arrow points from the box to "1800".



What is Data Science?

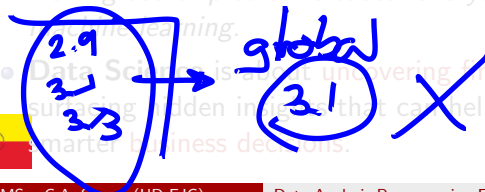
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garbage

ML

ML Ops

Data Ops



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CEO → Chief
CTO → Chief

COO → Chief
CAO → Chief

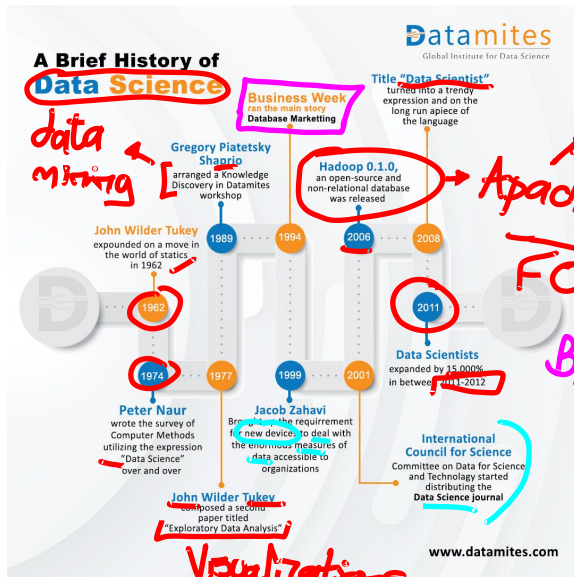


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- **Data Science** is about **uncovering findings from data**, and it's about surfacing hidden insights that can help enable companies to make smarter **business decisions**.



Brief History of Data Science





Data Systems & Big Data

~ 2005

- **Big Data** refers to extremely large data sets that may be analyzed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.
- **Data Systems** are the mechanisms to store, retrieve, and send data. They are crucial for handling big data.

SQL + NoSQL

2004

10 Gb

100

Big Data

500 Gb

Terabytes

2005

2013

Java

python

Rumpy



Data Systems & Big Data

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- **Data Systems** are the mechanisms to **store**, **retrieve**, and **send data**. They are crucial for handling **big data**.
- **Big Data** is characterized by 3Vs: **Volume** (amount of data), **Velocity** (speed of data in and out), and **Variety** (range of data types and sources).

Terabytes

1990
text
↓
format
table

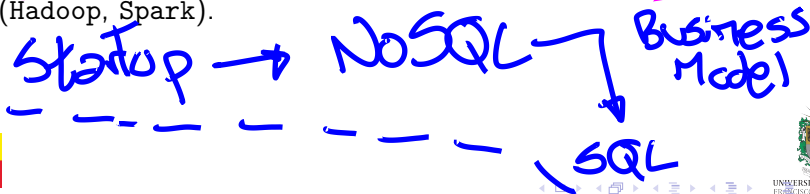
2000
images
video
sound
...

NoSQL
SOAP + XML

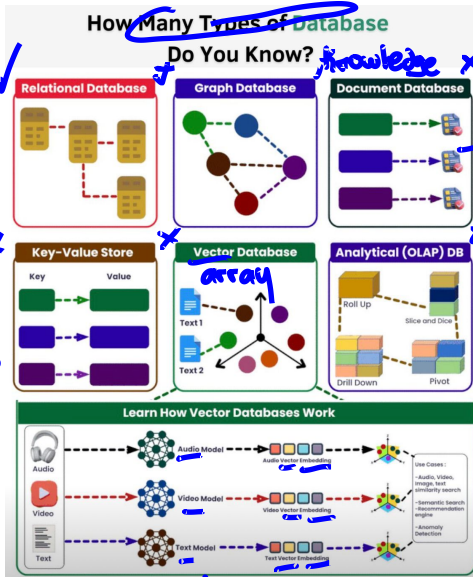


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- **Data Systems** for **Big Data** include **traditional databases** (SQL), **NoSQL databases** (MongoDB, Cassandra), and **distributed systems** (Hadoop, Spark).



Types of Database



JSON
Mongo

Dict
Apache
Redis

Real-Time

0.1GB
2GB

GPU → Chip AI



Data Lake Vs. Data Warehouse

- **Data Lake** is a storage repository that holds a vast amount of raw data in its native format until it is needed. It is a place to store every type of data in its native format with no fixed limits on account size or file.

Google Drive

OneDrive

files

txt

mpg

mp3

json

csv

word

pdf

excel

Original format



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/equivalent

- **Data Warehouse** is a system used for reporting and data analysis, and is considered a core component of business intelligence. It is a central repository of integrated data from one or more disparate sources.

Data Lakehouse is a hybrid approach that combines the features of both data lakes and data warehouses. It allows for the storage of raw data like a data lake while also providing structured and organized data capabilities of a data warehouse.

tables → relational databases

ML ← Query → Reports

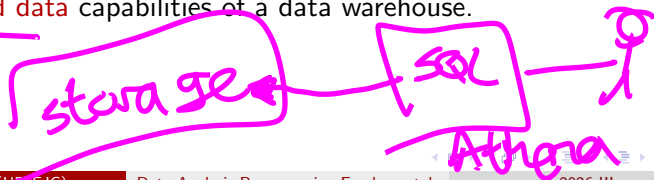


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cloud computing

Fin Ops



Data and MetaData boom

- **Data** refers to raw, unprocessed, and unorganized facts or details that alone might not make much sense or provide context.
- Metadata is data about data. It provides the *who, what, where, when, why, and how* of the data.
- Examples of metadata include *file size, creation date, modified date, and file type* for a digital file.
- Metadata helps in *data discovery, organization, and interpretation*.
- Metadata is crucial in data management practices like *data governance, data cataloging, and data lineage*.



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Data Governance



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↓
policies
+
docs

↓
searches
petabytes

data
versions

↓
tracking
changes



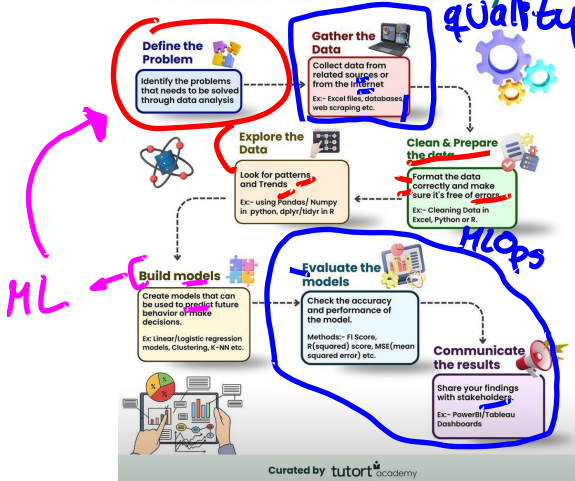
Process of Data Analysis



tutort.academy
#APlaceToGrow

Save for later

Process of Data Analysis



Types of Data Analysis

Different Types of Data Analysis



@alexmora



Descriptive Analysis

Summarizes basic data characteristics, like averages, frequencies, and distributions.

Answer "What happened?"



Diagnostic Analysis

Digs deeper, identifying factors influencing specific outcomes or trends.

Answer "Why did this happen?"



Predictive Analysis

Forecasts future events or values based on historical data patterns.

Answer "What might happen next?"



Prescriptive Analysis

Recommends optimal actions based on predicted outcomes and potential impacts.

Answer "What should we do now?"



Other Important Types That Are Widely Used

Exploratory Data Analysis (EDA)

Initial, open-ended investigation to understand data characteristics and uncover potential insights.

Inferential Analysis

Draws conclusions about a larger population based on a sample, using statistical methods to assess confidence levels.

Causal Analysis

Identifies factors causing specific outcomes, often using advanced techniques like causal inference methods.

Mechanistic Analysis

Examines underlying mechanisms and relationships within complex systems, often involving simulation or modeling.

patterns
+
trends

system
analysis

ML



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Artificial Intelligence for Data Science

- **Artificial Intelligence (AI)** is the simulation of human intelligence processes by machines, especially computer systems.
- **AI** involves learning, reasoning, and self-correction.
- **AI** is used in **data science** to *analyze data, make predictions, and automate decision-making processes.*
- **AI** techniques used in **data science** include *machine learning, deep learning, and natural language processing.*

↓
machines who
solve problems
as a human



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Data Science in Industry

- **Data Science** is used in **many industries** to **make decisions** optimize processes, and increase efficiency.
- Data Science is used in **healthcare** to *predict patient outcomes, optimize treatment plans, and personalize medicine.*
- Data Science is used in **finance** to *detect fraud, predict stock prices, and automate trading.*
- Data Science is used in **retail** to *optimize pricing, forecast demand, and personalize marketing.*
- Data Science is used in **manufacturing** to *predict equipment failures, optimize supply chains, and improve quality control.*
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Predictive
Prescriptive



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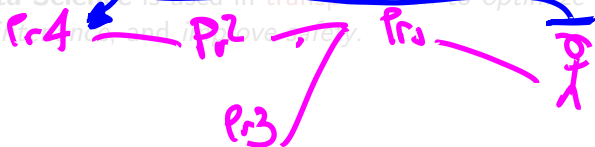
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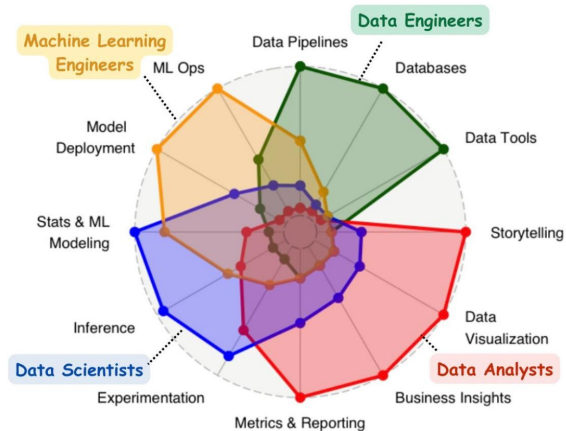
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Tech Team — Roles

Types of Data Roles - Where are you?



Tech Team — Data Profiles

WHICH PROFILE DESCRIBES YOU THE MOST?

ML ENGINEER



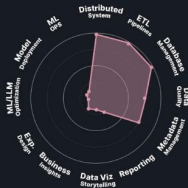
DATA STEWARD



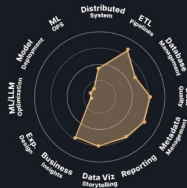
DATA SCIENTIST



DATA ENGINEER



ANALYTICS ENGINEER



DATA ANALYST



Data Scientist Responsibilities

- Collecting large sets of structured and unstructured data from **disparate sources**.
- Cleaning and validating the data to ensure **accuracy**, **completeness**, and **uniformity**.
- Analyzing the data to identify **patterns** and trends.
- Interpreting the data to discover solutions and **opportunities**.
- Communicating findings to stakeholders using **visualization** and other means.
- Developing, prototyping, and implementing **machine learning models**.
- Staying current on techniques and tools in the field, and continually **improving skills**.



Artificial Intelligence Tech Ecosystem

AI Infrastructure Tools open source

AI FRAMEWORKS, TOOLS & LIBRARIES



AI MODELS



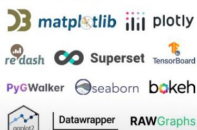
SEARCH



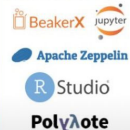
LOGGING & MONITORING



VISUALIZATION



COLLABORATION



@alexwang



Data Science Python Tech Ecosystem

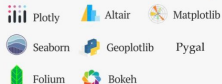
Life is Short, I Use Python



Data Manipulation



Data Visualization



Statistical Analysis



Machine Learning



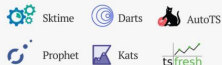
Natural Language Processing



Database Operations



Time Series Analysis



Web Scraping



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

